

Water Treatment Program Guidelines

Sego Industrial Water internal reference. Revision 6, February 2026. Prepared by D. Whitlock with field review by senior technicians.

1. Purpose and scope

This document defines control limits, interpretation rules, and escalation requirements for the treatment programs Sego operates at customer sites. It covers steam boilers, open recirculating cooling systems (towers and evaporative condensers), closed loops, and pretreatment equipment.

These are operating guidelines, not a substitute for judgment. Two systems with identical readings can deserve different responses depending on program, equipment, and history. Where this document and a system-specific program sheet disagree, the program sheet governs, and the technician should flag the disagreement for reconciliation.

2. Sampling and data recording standards

Consistent recording matters as much as consistent treatment. Standard units for Sego field records are:

Parameter	Standard unit
Conductivity	microsiemens per centimeter (uS/cm)
Hardness, alkalinity	ppm as CaCO ₃
Chloride, iron, nitrite, molybdate, sulfite, inhibitor	ppm
Microbiological activity	dipslide culture, reported as 10 ^x CFU/ml

Field meters, controller displays, and laboratory instruments do not all use these units. Conductivity in particular is commonly displayed in mS/cm on some controllers and lab reports (1 mS/cm = 1,000 uS/cm), and some technicians trained in other industries record hardness in grains per gallon (1 gpg = 17.1 ppm as CaCO₃). Values must be converted to standard units when

entered into customer-facing reports. Where a record carries its own unit notation, the notation governs over any assumption.

Samples should be drawn from designated sample points after an adequate flush. Samples from dead legs, stagnant branches, or pot feeder shells are not representative and should be re-drawn; if an unrepresentative result has already been recorded, it stays in the record and is annotated rather than deleted.

3. Control limits by treatment program

The limits below are the default bands for each program. Section 6 lists exceptions that override this table for specific system classes. Applying this table without checking Section 6 is the most common error new technicians make.

3.1 Steam boilers

Parameter	BP-STD (up to 150 psig)	BP-HP (150 to 400 psig)
Boiler water conductivity	2,200 to 3,500 uS/cm	800 to 1,500 uS/cm
Boiler water pH	10.5 to 11.5	9.8 to 10.5
M alkalinity	500 to 800 ppm	200 to 400 ppm
Sulfite residual	20 to 40 ppm	30 to 50 ppm
Feedwater hardness	below 2 ppm	below 1 ppm
Boiler water iron	below 2.0 ppm (action level, see 4.2)	below 1.0 ppm

3.2 Open cooling systems (towers and evaporative condensers)

Parameter	CT-STD	CT-HC2 (high cycles, see 6.1)
Conductivity	800 to 1,800 uS/cm	per 6.1
pH	8.0 to 9.0	8.2 to 9.0
Total hardness	below 750 ppm	per 6.1
Chloride	below 300 ppm	below 300 ppm
Free oxidizing biocide	0.5 to 1.0 ppm	0.5 to 1.0 ppm
Inhibitor residual (CT-770)	8 to 14 ppm	n/a
Inhibitor residual (CT-785)	n/a	12 to 18 ppm
Dipslide	below 10 ⁴ CFU/ml (see 5)	below 10 ⁴ CFU/ml (see 5)

3.3 Closed loops

Parameter	CL-N (nitrite)	CL-M (molybdate)
Nitrite residual	900 to 1,200 ppm	n/a
Molybdate residual	n/a	200 to 300 ppm
pH	8.5 to 10.5	8.0 to 10.0
Iron	below 1.0 ppm	below 1.0 ppm
Conductivity	record for trend; no fixed limit	record for trend; no fixed limit

A closed loop holding its inhibitor residual between services is healthy. A loop that repeatedly consumes inhibitor is telling you it is not closed: look for water loss before adding more chemical.

4. Interpreting results: excursions versus trends

4.1 Single excursions

A single out-of-band reading is a prompt to verify, not an emergency. Re-test where practical, check the sample point, check the unit notation, and check for a site event (rain dilution of open towers, recent makeup, recent blowdown). Physically impossible values (for example a pH outside the 0 to 14 scale, or a negative concentration) are recording or instrument errors by definition and must be corrected in the record, not acted on.

Open towers diluted by heavy weather will read low on conductivity and inhibitor for several days and recover on their own. Do not chase setpoints after a storm.

4.2 Sustained trends

A parameter moving steadily in one direction across three or more consecutive services indicates an active mechanism, and the time to investigate is at the trend stage, not at the limit. This applies even when every individual reading is inside its band. The canonical example: boiler or loop iron climbing month over month from a stable baseline indicates active corrosion well before the 2.0 ppm action level is reached, and waiting for the limit generally means the metal loss has already happened. Rising tower conductivity against a fixed bleed setpoint, falling biocide residual at constant feed, and growing gaps between chemical dose and residual all earn the same treatment: find the mechanism.

Trend judgments require consistent units across the series. Verify unit notation before concluding a step change is real; a controller or lab scale change can masquerade as a thousandfold excursion.

5. Microbiological control

5.1 Monitoring

Open recirculating systems receive a dipslide culture at every routine service. Results are semi-quantitative powers of ten.

5.2 General sites

Dipslide result	Response
10 ³ or below	Normal. Routine program.
10 ⁴	Corrective: verify oxidant residual and feed equipment, supplemental biocide, re-test at next service.
10 ⁵ or above	Elevated: corrective actions plus physical inspection for fouling; consider offline cleaning if regrowth recurs.

5.3 Healthcare and assisted-living sites (addendum)

Facilities serving patient or resident populations run a tighter ladder because of aerosol exposure risk. For these sites:

Weekly dipslides are required on all open towers regardless of visit frequency elsewhere in the contract. A result of 10⁴ requires same-visit corrective action and a re-test within one week. Two consecutive results at or above 10⁴ require notification of the Sego operations director within one business day, a documented remediation plan communicated to the customer in writing, and continued weekly re-tests until two consecutive results return to 10³ or below. Any single result at 10⁵ requires an urgent recommendation for offline cleaning and disinfection of the affected tower.

Verbal mentions to site staff do not satisfy the notification requirement in this addendum. The escalation exists on paper or it did not happen.

6. Program-specific exceptions and notes

6.1 CT-HC2 high-cycles cooling program

Systems formally enrolled in CT-HC2 (confirmed by the treatment_program field on the system record, not by observed chemistry) intentionally operate far above the standard conductivity band to reduce water consumption. For CT-HC2 systems the conductivity band is 2,400 to 3,400 uS/cm, provided both of the following hold: chloride remains below 300 ppm, and CT-785 inhibitor residual remains at or above 12 ppm. If either condition fails, the system reverts to CT-

STD limits until restored. Conductivity between 1,800 and 3,400 on an HC2 system is not an exceedance and must not be reported to the customer as one.

6.2 High-pressure boilers (BP-HP)

BP-HP units run much tighter conductivity than BP-STD. Readings that would be comfortably normal on a standard boiler are serious exceedances on an HP unit. Route coverage changes are the moment this mistake happens; check the program before judging the number.

6.3 Molybdate loops serving process tooling

For CL-M loops serving semiconductor or precision tooling, hold molybdate in the upper portion of the band, 225 to 300 ppm, because tool warranties reference that figure. The 200 ppm floor in Section 3.3 remains the contract minimum; between 200 and 225 is compliant but should be topped up at the same visit.

6.4 Towers on reuse or reclaim makeup water

Towers fed partially with reuse water will drift upward in cycles and conductivity as reuse fraction increases. This does not make them HC2 systems. The HC2 band applies only to formally converted systems with the required inhibitor and monitoring. For a reuse-fed tower drifting above the CT-STD band, the correct paths are: tighten bleed control, reduce reuse fraction, or formally convert the program after a chloride and scaling review. Quietly tolerating the drift is not one of the paths.

7. Seasonal layup and shutdown procedures

Sites with seasonal operation (resort properties, schools on summer schedules, seasonal agriculture) receive a documented layup at shutdown: boilers are laid up wet with elevated sulfite and buffered alkalinity or dry with desiccant depending on duration, loops are left circulating with inhibitor verified, and open towers being idled are drained and cleaned. Layup and recommissioning are recorded as service visits. A seasonal site showing no visits during its documented closure window is on schedule, not neglected; the distinction lives in the contract terms, and coverage reviews should read them before raising an alarm.

8. Products and typical consumption

Typical maintenance consumption for a healthy system, for budgeting and anomaly detection. Sustained consumption well above these figures without a program change is a finding, not a billing quirk.

Product	Application	Typical consumption
BW-210 sulfite blend	Boilers	1 to 3 gal per service
BW-305 alkalinity builder	Boilers	0 to 2 gal per service
AM-115 condensate amine	Boilers with condensate return	about 1 qt per service; do not skip
CT-770 inhibitor	CT-STD towers	3 to 6 gal per service
CT-785 inhibitor	CT-HC2 towers	3 to 6 gal per service
BIO-12 / BRM-40 biocides	Open systems, alternated	1 gal / 5 lb per service
CL-40 nitrite/borate	CL-N loops	initial charge about 1 gal per 250 gal volume; maintenance near zero on a tight loop
CL-M40 molybdate	CL-M loops	maintenance near zero on a tight loop

The closed-loop lines deserve emphasis: after initial charge, a tight loop consumes almost nothing. Recurring multi-gallon CL-40 or CL-M40 dosing at one site means the loop is losing water, and the chemical spend is the symptom, not the problem.

9. Escalation and documentation requirements

Any condition presenting a safety, compliance, or equipment-integrity risk must be captured as a work order within one business day of identification, regardless of whether site staff were told in person. Verbal notification is a courtesy; the work order is the record. Conditions that meet this bar include, at minimum: microbiological escalations under Section 5.3, suspected active corrosion under Section 4.2, treatment interruptions on critical systems, and any repeated equipment failure after a completed repair.

A work order closed after a repair that later recurs is reopened or re-raised with reference to the original; closing the paperwork does not close the problem.

Customer-facing commitments (a promised visit, a promised document, a promised plan) are tracked to completion by the service coordinator. A commitment made in email and not scheduled is a miss waiting to be discovered by the customer.

10. Outside laboratory reports

Central Analytical, Sego's contract laboratory, reports conductivity in mS/cm and identifies sample points with its own codes rather than Sego system IDs. Both conventions differ from Sego

field records. Lab results must be converted to standard units and mapped to the correct system before they are compared with field data. The current code crosswalk is maintained informally by the senior technician group; formalizing it is a known gap.

A lab result and a field result that disagree wildly usually differ by units, sample point, or timing. Resolve the mundane explanations before concluding the water changed.